Embedding geometry project notes

Oct 27, 2026 · @Talia

The project tests whether a language model's geometry is good enough for steering along an intensity scale (good < great < excellent), without assuming its representation space is a smooth manifold. It feeds a Substack post for The Hedge, a likely sequel on fine-tuning, and an email to Jon Rawski.

The question

The working question is now: is the geometry sufficient for steering along an intensity scale? It started as "is embedding space a smooth manifold?" and changed in three steps.

1. **The unstated assumption.** Steering vectors, UMAP readings, cosine-as-meaning and interpolation between activations all assume representation space is smooth without saying so.
2. **Smoothness can't be confirmed.** With finite samples you can only reject smoothness at some scale, never confirm it. And the network itself is smooth by construction (GELU, LayerNorm, attention), so a "snap" in outputs can only mean a very steep stretch, not a true discontinuity.
3. **The problem-sufficiency reframe.** From Levenstein, De Santo, …, Rawski & Wright, "The Problem-Ladenness of Theory" (2024): an idealization is judged by whether it is accurate enough for the problem it is used to solve. Smoothness is the idealization; steering is the problem; steering only needs something weaker, order preserved along a path.

One-sentence versions:

- **For anyone:** testing whether moving a model's internal state along an "intensity" direction makes its output more intense smoothly and predictably (good → great → excellent), or whether it jumps or breaks.
- **Technical:** testing whether intensity scales like good < great < excellent are encoded as a direction you can actually steer along: monotone, graded and coherent.
- **Thesis line:** testing whether a language model's geometry is good enough for steering, without assuming it is a smooth manifold.

Embedding geometry, for someone who took linear algebra: each token becomes a vector in a space of roughly 768 to 4,000 dimensions, updated layer by layer, and embedding geometry is how those vectors are arranged (distances, directions, clusters) and what that says about meaning. A word maps to a whole vector, not to a dimension.

What the probe tests

Steering along an intensity direction succeeds if four criteria hold. A smooth manifold with intensity as a coordinate would guarantee all four; the claim under test is that only these four are needed.

Criterion	What success looks like	Test that covers it	Status
Monotone	A bigger steering coefficient gives a more intense output, never a reversal	Probe readout along the interpolation path; direct steering sweep	Readout built; sweep not built
Graded	An intermediate coefficient gives an intermediate intensity (a dial, not a switch)	Step-KL plus the resolution ratio	Built
Near the data	Steered activations stay close enough to real ones that output stays fluent	Distance from the nearest real activation, relative to typical spacing	Nearest-word only; distances not built
General	The same direction works across contexts and on words it wasn't built from	Leave-one-word-out probe across frames	Built

Monotonicity is the spine. On its own it is too weak: a step function is monotone but can't be dialed, and intensity that keeps rising while output turns to gibberish is monotone but a failed steer.

The current probe runs three tests:

1. **Order.** Is there a linear direction that ranks the scale, including a word the probe never saw, clearly better than a random-weights model?
2. **Arrangement.** Is the ordered path through the word centroids straighter than random orderings of the same points, and does PC1 recover the order beyond chance?

3. **Traversal.** Mixing *good* and *excellent* at layer L, does the output change gradually, and does the path pass near *great*?

Outcomes that each make a post:

- **All pass:** steering on this scale is geometrically justified at those layers; the manifold assumption was overkill.
- **Order passes, traversal fails:** the scale is decodable but not traversable. Probably the most interesting result.
- **Order fails as badly as the random model:** no scale direction; earlier decoding was token identity.

Gap: the probe interpolates between two words' activations, which is a proxy. Real steering adds λ·v to an arbitrary context. A direct sweep (add λ times the intensity direction to neutral sentences, plot output intensity and fluency against λ) tests the four criteria on the actual operation and should go in before the multi-model runs.

The smoothness probe (smoothness_probe.py, v2)

The script runs end to end but has only been tested on a tiny randomly initialized stand-in model (4 layers, 64 dimensions), because the model hub was blocked from the sandbox. No real-model numbers exist yet.

Setup: TransformerLens 3.x (pin `transformer_lens<4`; 4.0 removed HookedTransformer), loaded with `from_pretrained_no_processing`. It runs as notebook cells or top to bottom; all settings sit in a config block at the top.

Inputs

- Word set (default, one pole): okay, decent, good, great, excellent, amazing, perfect. A two-pole valence set (terrible … perfect) remains as a switch, to be reported as a sentiment scale. Multi-token words are dropped automatically; check the kept/dropped printout per model.
- 12 sentence frames ("The movie was", "Honestly, the food was", …). Frame 0 is the interpolation frame, and probes never train on it.
- Pairs to interpolate: good → excellent, okay → perfect, and good → purple as a control across the semantic gap.

Experiment 1: geometry, per layer

- PC1 of frame-centered activations vs rank: |Spearman| plus a permutation p-value from 1,000 shuffled orderings.

- Tortuosity (path length through ordered word centroids divided by endpoint distance), reported as a percentile against shuffled orderings.
- Anisotropy: mean pairwise cosine of raw activations.
- Leave-one-word-out ridge probe: train on all words but one, predict the held-out word's rank; Spearman and p-value across held-out predictions.
- PCA plots at five depths (the hero figure).

Experiment 2: interpolation

- Norm-matched mixing of word A's and word B's activations at a layer, patched into the last position, then run to the end.
- Readouts: a final-layer probe trained without the pair's endpoint words; step KL over the full next-token distribution and over content tokens only; the nearest real word at each step.
- Resolution scaling: 41 vs 161 steps. A max-KL ratio near 16 means smooth at this resolution; near 1 means a transition narrower than the fine step.
- A last cell repeats one pair without norm matching to show the norm-dip artifact.

Guards

- The scalar word is always the last token, so patching can't leak word A through lower layers.
- An assert that $\alpha = 0$ reproduces the clean run for word A.
- A RANDOM_BASELINE switch (same tokenizer, re-initialized weights).
- When the patch layer is the final layer, the probe readout is linear by construction; it is plotted dotted and not counted.
- Float64 log-softmax so fine-resolution KLs stay above numerical noise.

Test-run notes (tiny model only)

- Every pair gave a resolution ratio of 16.0, as expected for a smooth random model.
- The random model's leave-one-word-out ρ reached about 0.6–0.7 at $p \approx 0.1$. With 7 words, chance can look like signal; more words may help more than any other fix.

Review feedback on v1

An outside review of v1 was right on nearly every point; v2 adopts all of it except the digit readout.

Point	Verdict	Change in v2
Probe held out over frames, not words, so it can memorize token identity → rank	Correct	Leave-one-word-out probe
Max/median KL can't tell a discontinuity from a smooth but uneven path	Correct	41 vs 161 steps; report the max-KL ratio (≈16 smooth, ≈1 sharp)
Tortuosity and PC1 Spearman are uninterpretable without a null	Correct	1,000-shuffle nulls; percentile and p-value
terrible … perfect is a valence scale, not an intensifier scale	Correct	One-pole default list; valence kept as a labeled switch
The chord may run off the data	Correct	Nearest real word at each step; distances still to build
Assert α = 0 matches the clean run	Correct	Added
Final-layer patch readout is linear by construction	Correct	Marked trivial, plotted dotted
Default weight processing (centering) distorts geometry	Correct	from_pretrained_no_processing
Full-vocabulary KL is dominated by punctuation; use an "out of ten" digit readout	Half right	Content-token KL instead. A suffix after the word reads its unpatched lower layers and leaks word A
sweep() depends on a variable from an earlier cell	Correct	Config moved to the top
The random model will fail leave-one-word-out	Not confirmed	On the tiny model it reached ρ ≈ 0.65 (p ≈ 0.1); script now reports p-values

Limits: what the probe assumes, and what a manifold test would take

The probe tests order and monotonicity, not manifold-ness. It assumes a vector space (mixing, PCA, linear probes), a Euclidean metric (tortuosity, nearest word, cosine), and differentiability of the network, which is free.

- **The metric is the weakest assumption.** A metric is more structure than a topology, and GPT-2's anisotropy and rogue dimensions can dominate Euclidean distance (Timkey & van Schijndel 2021). A whitening or cosine switch would check whether results survive.
- **The one intrinsic measure is step KL.** KL between nearby output distributions is the Fisher metric pulled back onto the residual stream: the distance the model itself is sensitive to.
- **Not tested:** whether activations lie on a manifold, any topology, whether the model "represents intensity" in a deeper sense, anything beyond the models, layers, words and metric used.

A real manifold test needs a point cloud of thousands to tens of thousands of activations (e.g., every token position over ~10k sentences at one layer), and a decision about which object is tested: the token embedding table or contextual activations.

Layer	Math	What it checks	Tools / references
1. Intrinsic dimension	Statistics	Local dimension is constant across neighborhoods	TwoNN (Facco et al. 2017), Levina–Bickel MLE; scikit-dimension
2. Local flatness, tangent spaces	Linear algebra	Neighborhoods are flat d-planes; nearby tangent planes nearly parallel (principal angles)	Local PCA
3. Scale and curvature	Differential geometry	Reach: the largest scale at which the set still looks like a manifold; volume growth ~ r^d	Niyogi–Smale–Weinberger; Fefferman–Mitter–Narayanan; Robinson et al. 2025
4. Topology proper	Algebraic topology	Local homology: the ring around each point looks like a sphere; singular points fail	ripser, giotto-tda; persistent local homology (Bendich et al.)

A version that keeps the scalar-adjective thesis needs only layers 1–2: collect activations for good / great / excellent across thousands of natural contexts, estimate local dimension and tangent planes, and ask whether the intensity direction is tangent to the data or points off it. If it points off, steering along it leaves the data by construction.

Microlocal homology. The paper found under that name (Schefers, arXiv 2205.12436, now "An Equivalence Between Vanishing Cycles and Microlocalization") is derived algebraic geometry: invariants of singular spaces indexed by sets of covectors (directions), interpolating between cohomology and Borel–Moore homology. It is purely theoretical, set in complex analytic spaces, with no estimator for point clouds. It fits as a lens: Robinson et al. detect singularities but say a rejection doesn't identify their type, and a microlocal description would classify them by direction. Use it as a footnote framed as an open problem. The computable neighbors are persistent local homology and Kashiwara & Schapira's "Persistent homology and microlocal sheaf theory."

Models and the two posts

GPT-2 is a smoke test only. It was the default because it is cheap, best supported in TransformerLens and common in the literature, but it is old, small and unusually anisotropic. The experiment is tiny (inputs of about ten tokens), so 1–2B models run in minutes on a recent Mac or free Colab.

Post 1: does it hold across models?

Model	Why	Runs on
Pythia 160M / 410M / 1.4B (→ 2.8B)	Size sweep on identical data	Laptop
Pythia training checkpoints (optional)	When the scale emerges; step 0 is a principled random baseline	Laptop
Llama-3.2-1B	Modern, widely steered; gated on Hugging Face	Laptop / free Colab
Qwen2.5-1.5B or Gemma-2-2B	A different family; Qwen is ungated	Laptop / free Colab

Post 2: does fine-tuning move it?

Model	Role
Llama-3.1-8B	Base
Llama-3.1-8B-Instruct	Chat fine-tune
DeepSeek-R1-Distill-Llama-8B	Reasoning fine-tune of the same base

The 8B models need a ~16GB GPU (a free Colab T4 is at the edge; a rented A100 for an afternoon costs a few dollars). The distill probably loads via `hf_model=` with the Llama architecture name; check before committing.

Post 2 adds a transfer test: the cosine between the intensity direction found on the base model and on each fine-tune, and whether the base model's steering vector still works on each fine-tune. Steering vectors are often found on one checkpoint and deployed on another.

Not possible:

- **Claude:** closed weights and, as far as known, no API log-probabilities. Only a behavioral section (forced choice, graded ratings) is possible, clearly labeled as behavior, not geometry.
- **Full DeepSeek V3 / R1:** 671B-parameter mixture-of-experts; only the distills are in reach, and those are Qwen or Llama underneath.

Order: finish post 1 first, so the harness is debugged on cheap models before paying for GPU time.

Prior work and what is new

That an intensity direction exists has been shown before. What hasn't turned up in searching is whether a graded scale direction can be traversed, how that changes with scale and training, and whether it survives fine-tuning.

Work	Finding	Relation to this project
Garí Soler & Apidianaki 2020	An intensity direction built from a few adjective pairs ranks held-out scalar adjectives in BERT	Closest prior work to the leave-one-word-out probe; encoder, not decoder residual stream. Add their difference-vector method as a baseline
Lin, Altshuler & Pierrehumbert 2024	LLMs encode scalar adjective intensity but handle scalar implicature poorly	Home literature; read it
Tigges et al. 2023	A single linear sentiment (valence) direction in GPT-2 and Pythia, with causal interventions	The prior result readers will ask about; valence, which the one-pole list avoids
Mishra, Khashabi & Liu 2026	Steered activations almost surely cannot be produced by any prompt	The chord is off the reachable set exactly; report distances, not membership

Work	Finding	Relation to this project
Robinson, Dey & Chiang 2025	Volume-growth test rejects the manifold hypothesis at specific tokens in GPT-2 and three 7B models	Token embeddings only; says a rejection doesn't identify the singularity's type
Rimsky et al. (CAA, Llama 2) and work on base-to-chat SAE transfer	Steering vectors and SAEs reportedly transfer from base to chat models	Context for the post 2 transfer test; nothing found on reasoning distills

Other references used: Timkey & van Schijndel 2021 (rogue dimensions), Ethayarajh 2019 (anisotropy), Elhage et al. (superposition), Ansuini et al. and Valeriani et al. (intrinsic dimension across layers), Park, Choe & Veitch (linear representation hypothesis), de Melo & Bansal 2013 (intensity ratings).

What is new here:

- Traversability of a graded, one-pole scale (interpolation, resolution scaling, distance from data), not just decodability
- Scale and training dynamics of that direction
- Survival under reasoning distillation, and base-vector transfer
- Grounding the scale in entailment (Horn scales) rather than crowd ratings

The search was not exhaustive. Before publishing, spend an hour on "scalar adjective" with "steering" or "activation patching", 2024–26.

Jon Rawski

The email goes as a reply in the May 2026 thread, no greeting, signed "Talia", framed around his own paper.

Context

- Took his NLP course at SJSU in spring 2026.
- May 2026 thread: he read The Hedge; framed Marr's levels as "coarser/finer lenses" rather than a hierarchy; said ML theory is hitting the crises neuroscience hit around 2015; on May 20 sent "The Problem-Ladenness of Theory" in response to a point about reducing messy phenomena to utility functions. Unknown whether there was a reply after May 20; the email's opener works either way.

- His work: tensor product representations of formal languages (discrete, algebraic, no continuity claims); "Modern Language Models Refute Nothing" with Baumont (fit isn't theory; correlation can't establish that a model is a theory); BBS commentary with Heinz on Futrell & Mahowald; a 2025 SJSU grant, "Geometric Structures in Natural Language", with Edgar Bering (output not found).
- "Learning with Partially Ordered Representations" orders relational structures by embedding, not meanings; don't cite it as related to intensity.
- Strongly anti AI consciousness: avoid "knows", "understands", "wants", and keep "represents" tied to what a probe can recover.

Final email

> I've been meaning to come back to the problem-ladenness paper you sent, and it ended up reorganizing a project I'm working on.
>
> A lot of interpretability practice assumes LLM representation spaces are smooth manifolds without saying so: steering vectors, UMAP readings, interpolating between activations. I'd been framing my question as "is the space actually smooth?" But with finite samples and a network that's smooth by construction, that's close to unanswerable. Your paper made me see it as a problem-sufficiency question instead. Smoothness is an idealization, and what something like steering actually requires is weaker: that the geometry preserves order along a path. So I'm planning to test whether activations preserve a Horn scale (good < great < excellent) across a few open models. I'll use a leave-one-word-out probe against a random-weights baseline, since a probe trained on every word can recover the ordering from word identity alone.
>
> (Box 3's Hopfieldian view also reads like the neuro ancestor of the manifold assumption, which made your comment about ML hitting neuro's 2015 crises click for me.)
>
> Does that seem like the right way to cash out the paper here, or am I misapplying it?
>
> Talia

Next steps

- [ ] Send the email to Jon as a reply in the May thread
- [ ] Run v2 on GPT-2 as a smoke test (pin transformer_lens<4)

- [ ] Add the direct steering sweep (λ·v on neutral sentences; intensity and fluency vs λ)
- [ ] Build the multi-model harness: model list, saved results per model, relative-depth plots, hf_model= loading
- [ ] Replace nearest-word with distance from a bank of real activations, relative to typical spacing
- [ ] Add a Garí Soler–style difference-of-means baseline direction
- [ ] Add a whitening / cosine switch for rogue dimensions
- [ ] Ground the word list in human intensity ratings; add more words if the random baseline isn't clearly beaten
- [ ] Run post 1: Pythia sweep, Llama-3.2-1B, Qwen2.5-1.5B or Gemma-2-2B
- [ ] Add the transfer test and run post 2 on a rented GPU
- [ ] One hour of literature search, 2024–26, before publishing
- [ ] Ask someone with geometry or topology background to read the draft

Separate from this project: the OpenReview profile for the SPAR paper is confirmed. An arXiv account isn't needed to submit; ask Varad whether the team posts a preprint and who submits it.

Sources

- The Problem-Ladenness of Theory (Levenstein et al., Computational Brain & Behavior 7:548–571, 2024), PDF shared by Jon Rawski
- Modern Language Models Refute Nothing (Rawski & Baumont)
- You Can't Be Neutral on a Moving Train (Heinz & Rawski)
- Tensor Product Representations of Subregular Formal Languages
- Tensor Product Representations of Regular Transductions
- Learning with Partially Ordered Representations
- BERT Knows Punta Cana is not just beautiful, it's gorgeous
- Probing LLMs for Scalar Adjective Lexical Semantics and Scalar Diversity Pragmatics
- Linear Representations of Sentiment in Large Language Models
- Steered LLM Activations are Non-Surjective
- Token Embeddings Violate the Manifold Hypothesis
- An Equivalence Between Vanishing Cycles and Microlocalization (Schefers)